

Literature Review

Causal RL with hidden confounders and partial observability

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1 Selected Papers

The following table lists the core papers selected for this literature review, focusing on the intersection of partial observability and unobserved confounders in offline reinforcement learning.

#	Paper	Venue / Year	Why it is relevant
1	Model-based Reinforcement Learning for Confounded POMDPs	ICML, 2024	The core paper of our project that uses a model-based approach and pessimism to handle confounders.
2	Pessimism in the Face of Confounders: Provably Efficient Offline Reinforcement Learning in Partially Observable Markov Decision Processes	arXiv, 2024	A great model-free alternative that also uses pessimism, making it perfect for comparing methods.
3	A Minimax Learning Approach to Off-Policy Evaluation in Confounded Partially Observable Markov Decision Processes	arXiv, 2021	Solves the problem in continuous spaces using a minimax learning method instead of tabular approaches.
4	Off-Policy Evaluation in Partially Observable Environments	AAAI, 2020	A foundational paper that clearly explains why traditional offline evaluation fails in POMDPs.
5	Proximal Reinforcement Learning: Efficient Off-Policy Evaluation in Partially Observed Markov Decision Processes	arXiv, 2021	Introduces causal inference and proxy variables to bridge the gap caused by hidden confounders.
6	A Spectral Approach to Off-Policy Evaluation for POMDPs	arXiv, 2021	Uses linear algebra and spectral learning to uncover hidden dynamics without strong causal assumptions.
7	Confounding-Robust Policy Evaluation in Infinite-Horizon Reinforcement Learning	NeurIPS, 2020	Focuses on finding safe upper and lower bounds for a policy's value instead of trying to find an exact number.
8	Deconfounding Reinforcement Learning in Observational Settings	arXiv, 2018	A classic that uses deep learning and latent variable models to predict the unobserved confounders.

2 Paper Summaries

2.1 Paper 1: Model-based Reinforcement Learning for Confounded POMDPs

Summary: This paper serves as the primary foundation of our project. It introduces a novel model-based offline reinforcement learning algorithm designed specifically for Partially Observable Markov Decision Processes (POMDPs) where the data is affected by unobserved confounders. Unlike other methods that try to estimate the policy's value directly, this paper attempts to extract information about the actual reward and transition dynamics from the offline data. It achieves this by introducing reward-emission and dynamic-emission "bridge functions" (e.g., $\mathbf{b}_R$ and $\mathbf{b}_P$) to solve a sequence of conditional moment restrictions, such as $\mathbb{E}[R - \mathbf{b}_R(O, A) \mid Z, A] = 0$, using observable proxies Z . Finally, by applying a pessimistic penalty to the estimated value function—meaning the agent acts cautiously in states with high uncertainty—the authors provide theoretical finite-sample bounds. This mathematically proves that an optimal policy can be safely learned even when the offline dataset only has partial coverage.

Justification: We selected this paper because it establishes the absolute core theoretical framework of our work. Understanding its unique model-based identification and two-stage estimation process is essential for building our proof of concept. What makes this paper stand out for us is that its mathematical bounds do not restrict the size of the policy space, which is a common and severe limitation in model-free alternatives. It directly drives our main research question, which investigates the efficiency, safety, and mathematical guarantees of model-based causal RL in highly confounded environments.

2.2 Paper 2: Pessimism in the Face of Confounders: Provably Efficient Offline Reinforcement Learning in Partially Observable Markov Decision Processes

Summary: This paper addresses the exact same problem—finding an optimal policy from confounded offline POMDP data—but approaches it from a completely model-free perspective. It proposes the Proxy variable Pessimistic Policy Optimization (P3O) algorithm. Instead of trying to learn the environment’s underlying transition rules, P3O uses proximal causal inference and minimax estimation of the form $\min_h \max_g \mathbb{E}[\mathcal{L}(h, g)]$ to directly estimate the value of actions. It builds coupled “pessimistic confidence regions” around these estimates to counteract both confounding bias (hidden variables) and distributional shift (lack of data). Ultimately, it proves a solid $\mathcal{O}(n^{-1/2})$ suboptimality bound under a partial coverage assumption, where n is the number of trajectories.

Justification: This paper was chosen as the primary theoretical competitor and baseline to our core paper. Contrasting this model-free method with the model-based approach of Paper 1 is crucial for a strong critical analysis. Understanding how P3O directly estimates values via minimax loss gives us the perfect contrasting perspective to defend our specific algorithmic choices during the mentor review. It provides the necessary context to scope a mathematically rigorous “Model-Based vs. Model-Free” comparison for the upcoming project phases, significantly enriching our final discussion.

2.3 Paper 3: A Minimax Learning Approach to Off-Policy Evaluation in Confounded Partially Observable Markov Decision Processes

Summary: This paper evaluates policies in continuous environments where a hidden state acts as an unmeasured confounder, secretly influencing the logged actions and rewards. Since the evaluation policy only observes recorded observations, standard RL estimators become heavily biased. To adjust for this bias, the proposed setup uses the agent’s observable history as a mathematical anchor. Because hidden states make true importance weights and Q-values uncomputable, the authors define “bridge functions” to project these ideal values onto observable data. These functions are estimated in continuous spaces using a minimax game: one neural network learns the bridge function, while an adversarial discriminator network maximizes the projected Bellman error.

Justification: Combining these bridge functions into a Doubly-Robust estimator yields semiparametric efficiency—converging to the true policy value at the optimal theoretical rate with minimum variance. This unlocks the safe, mathematically guaranteed use of deep learning for policy evaluation in complex, confounded continuous environments.

2.4 Paper 4: Off-Policy Evaluation in Partially Observable Environments

Summary: This work addresses the off-policy evaluation (OPE) problem (finding V^{π_e}) in POMDPs. The authors demonstrate that this setting is inherently different from regular MDPs, highlighting how ignoring unobserved confounding variables can lead to evaluating suboptimal policies as optimal ones. They resolve this issue under two distinct setups: 1) The environment state u is unobservable, but an observation z depends on u such that the dimensionality of z is greater than or equal to that of u ($|z| \geq |u|$). The behavior policy selects actions entirely based on u , whereas the evaluation policy selects actions based on the observable history. In this setup, the expected reward is calculated by using past and future observations as proxies to estimate the value function. 2) A “Decoupled POMDP” where the state consists of an unobservable state u and an observable state z . The behavior policy acts based on both u and z , and an additional observable variable o is emitted independently by u . This allows the evaluation policy to use the history of both z and o , enabling value function estimation under much more flexible mathematical assumptions.

Justification: This paper was chosen because it lays the essential groundwork for OPE in POMDPs. It presents the very first exact OPE result for general POMDPs by generalizing static causal tools to dynamic RL. This literature review would be incomplete without the inclusion of this foundational work.

2.5 Paper 5: Proximal Reinforcement Learning: Efficient Off-Policy Evaluation in Partially Observed Markov Decision Processes

Summary: Building upon the challenges of OPE in POMDPs, this paper generalizes the setup for continuous, complex environments. Acknowledging that discrete mathematics cannot be solely relied upon, the authors assume access to two observable proxies for the unobservable true state at any given time step: an action-side proxy and an outcome-side proxy (e.g., the previous and current observations). These proxies must be conditionally independent given the hidden true state. To solve the evaluation problem, the paper adapts “Proximal Causal Inference”—a statistical method that uses two imperfect proxies to cancel out the bias of a hidden confounder—into the sequential RL setting. It uses the data from the two proxies to learn “bridge functions”, which act similarly to inverse propensity scores and Q-functions. It then demonstrates how to combine these learned bridge functions into a Doubly Robust estimator.

Justification: This paper is essential because it introduces the “Proximal Reinforcement Learning” framework, making it the first to successfully extend Proximal Causal Inference to evaluate dynamic, adaptive policies. Crucially, it moves the field past

restrictive tabular models by providing the first OPE estimators mathematically proven to be consistent, asymptotically normal, and semiparametrically efficient in continuous, confounded environments.

2.6 Paper 6: A Spectral Approach to Off-Policy Evaluation for POMDPs

Summary: This paper evaluates a policy’s value using observable observations as proxies for the hidden state via spectral and moment-matrix algebra, techniques borrowed from HMM and PSR learning. It replaces the restrictive requirement that observation and state spaces be equal-sized and invertible with a weaker rank condition, utilizing multi-step histories and futures as richer proxies. Ultimately, it identifies the policy value through linear-algebra proxies without the need for explicit model fitting.

Justification: This work was chosen because it solves the same OPE-under-confounding problem as the main paper but utilizes a completely different mathematical toolkit (spectral/moment matrices), providing a strong contrast and making the main paper’s bridge-function approach more legible.

2.7 Paper 7: Confounding-Robust Policy Evaluation in Infinite-Horizon Reinforcement Learning

Summary: This paper acknowledges that the exact policy value usually cannot be point-identified under hidden confounding. Instead, it computes sharp upper and lower bounds using a sensitivity model parameterized by $\Gamma \geq 1$, which bounds the maximum strength of the unobserved confounding. A larger Γ results in wider bounds, providing a guaranteed evaluation interval rather than a precise but potentially biased scalar estimate.

Justification: Chosen as the capstone for robustness methodology: it teaches the robustness mindset. When point identification fails, computing guaranteed bounds via a sensitivity parameter Γ serves as a rich source of practical ideas for the proof-of-concept phase.

2.8 Paper 8: Deconfounding Reinforcement Learning in Observational Settings

Summary: This paper learns a neural latent-variable model (a VAE with LSTM encoders) that reconstructs the hidden confounder from observational trajectories, then uses Pearl’s *do-calculus* to compute a deconfounded reward for an Actor-Critic architecture. The necessity of the *do-operator* arises because observational conditioning ($\mathbb{P}(Y \mid X)$) differs fundamentally from causal intervention ($\mathbb{P}(Y \mid do(X))$). For instance, observationally, carrying a lighter is highly correlated with lung cancer ($\mathbb{P}(\text{cancer} \mid \text{lighter}) > 0$). However, mechanically intervening to give someone a lighter does not change their actual cancer risk ($\mathbb{P}(\text{cancer} \mid do(\text{lighter})) = \mathbb{P}(\text{cancer})$). The algorithm explicitly models this hidden confounder to intervene correctly from observational data.

Justification: Chosen as the intuitive, historical on-ramp: it is the first paper linking confounding to full RL and teaches the core causal vocabulary (do-operator, proxies) via familiar neural network architectures.

3 Comparison and Discussion

A review of the selected literature reveals several diverging paths in how researchers tackle the intersection of partial observability (POMDPs) and unobserved confounders. We can compare these works across three main dimensions: the primary goal of the algorithm, the mathematical mechanism used to handle confounders, and the approach to learning the environment’s dynamics.

3.1 Off-Policy Evaluation (OPE) vs. Policy Optimization

The majority of the foundational work in this area has focused strictly on Off-Policy Evaluation (Papers 3, 4, 5, 6, and 7). These methods attempt to accurately estimate the value of a given target policy using historical, confounded data. While OPE is a necessary first step, it does not inherently teach the agent how to act optimally. Papers 1, 2, and 8 bridge this gap by moving from evaluation to full Policy Optimization. Notably, Papers 1 and 2 achieve this by integrating the principle of “pessimism” into their algorithms, which prevents the agent from overestimating the value of actions that are poorly represented in the offline dataset.

3.2 Mechanisms for Deconfounding

How do these algorithms actually remove the bias caused by hidden variables? The literature provides four distinct strategies:

- *Proxy Variables & Bridge Functions:* Papers 1, 2, 3, and 5 rely heavily on proximal causal inference. They use observable proxies (such as historical or future observations) to construct mathematical “bridge functions” that implicitly cancel out the confounding bias without needing to know the exact hidden state.

- *Explicit Latent Modeling*: In contrast, Paper 8 takes a deep learning approach. It uses Variational Autoencoders (VAEs) to explicitly guess and reconstruct the hidden confounders, and then applies Pearl’s *do-operator* to calculate rewards.
- *Spectral Learning*: Paper 6 avoids strong causal assumptions entirely. Instead, it uses linear algebra and moment matrices to uncover the hidden dynamics of the environment based solely on the observable data.
- *Sensitivity Analysis*: When exact deconfounding is impossible due to a lack of good proxies, Paper 7 offers a fallback strategy. Instead of finding an exact policy value, it calculates safe upper and lower bounds (robustness) based on how strong the confounding effect might be.

3.3 Model-Based vs. Model-Free Approaches

For the ultimate goal of finding the optimal policy, Papers 1 and 2 present a fascinating theoretical rivalry. Paper 2 (P3O) introduces a completely model-free approach, arguing that one can optimize a policy using minimax estimation directly on the value functions, bypassing the need to model the complex environment. However, our core paper (Paper 1) argues for a model-based approach. By explicitly learning the transition dynamics and reward models under the hood of the bridge functions, the model-based approach can provide stronger theoretical guarantees and better interpretability, making it the central focus of our continued research.

4 Possible Project Direction

Our review exposes a clear asymmetry: the theory around confounded POMDPs is rich, but it is almost entirely unvalidated and several of its assumptions remain fragile. Our core paper (Paper 1) is, notably, a purely theoretical work with **no experiments at all**—the authors explicitly leave a practical implementation as future work. This gives our project a natural mandate: close the gap between the theory and a working, empirically-tested system, and then push on the assumptions that the theory currently takes for granted.

4.1 Open Questions Not Yet Addressed

- *When does model-based beat model-free?* Papers 1 and 2 are framed as rivals with different guarantees—Paper 1’s bound is independent of the policy-class size $|\Pi|$, while Paper 2 (P3O) carries a $\sqrt{\log |\Pi|}$ factor but may satisfy a weaker existence condition. Paper 1 sketches the regime where each should win (small vs. large policy/reward spaces), but **no one has empirically tested this trade-off**.
- *What happens when the identification assumptions silently fail?* The completeness assumption and the existence of bridge functions are untestable from data. Papers 1, 2, 3, and 5 all collapse if these fail, yet none provides a diagnostic or a graceful fallback. Paper 7 offers robustness, but in a different setting (infinite-horizon, reward on the observed state), so its safety net does not directly transfer.
- *The identified reward distribution is never used.* Paper 1 identifies the full marginal reward distribution $p^\pi(r_t)$, not just its mean, but then optimizes only the expected return. No paper exploits this for risk-sensitive, distributional, or Bayesian RL.
- *Two notions of uncertainty are never unified.* Papers 1 and 2 quantify uncertainty from *finite samples* (confidence regions); Paper 7 quantifies uncertainty from *confounding strength* (Γ). These orthogonal sources of error have not been combined into a single guarantee.

4.2 Implementation Directions

- *Implement the core paper.* Build the two-stage RKHS (kernel ridge) estimator for the reward-emission and dynamic-emission bridge functions, plus the pessimistic max–min policy optimization, and reproduce it on a small synthetic confounded POMDP. To our knowledge, there is no public implementation, so this alone is a meaningful contribution.
- *Head-to-head: model-based (Paper 1) vs. model-free (Paper 2).* Implement P3O on the same environment and the same data, then sweep the policy-class size, the reward/state cardinality, and the dataset coverage to empirically locate the crossover point that the theory only predicts.
- *A reusable confounded-POMDP benchmark.* Extend the confounded Pendulum/CartPole/MNIST tasks of Paper 8 (and/or a sepsis-style task as in Paper 5) into a standard offline benchmark with controllable confounding strength and coverage—enabling all of the comparisons above.
- *Ablations.* Study how the quality and dimension of the baseline covariate O_0 (Assumption 3.1 in Paper 1), and the choice of kernel/regularization, affect accuracy and sample efficiency.

- *Cheaper estimation via spectral tools.* Test whether the moment-matrix / spectral machinery of Paper 6 can estimate the bridge functions of Paper 1 more cheaply or under weaker rank conditions, and compare against the kernel approach.

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